GH3036 & GH3038 EVK Quick Start Guide

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Shenzhen Goodix Technology Co., Ltd.

Headquarters: 26F, Goodix Headquarters, No.1 Meikang Road, Futian District, Shenzhen, China; Zip Code:518000

TEL: +86-755-33338828

Website: http://www.goodix.com

Preface

Purpose

This document is a quick start guide of GH3036 & GH3038 Health Evaluation Kit (hereafter referred to as EVK).

Audience

This document is intended for:

- Users of GH3036 & GH3038
- Software engineers
- Test Engineers

Release Notes

This document is the second release.

Revision History

Revision	Date	Description
0.1	2024-12-11	Preliminary release
0.2	2025-03-03	Updated the document format

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1 Overview

This user guide describes the operation of the EVK, which is an evaluation module for the GH3036 & GH3038 sensors.

GH3036 & GH3038 are dedicated for optical AFEs, which can be used for Health sensors. GH3036 supports 2 RX channels, maximum 8 PPG slots + 1 CAP slot. GH3038 supports 1 RX channel, maximum 8 PPG slots, but it does not support capacitance measurement. The functions supported include HR, HRV, SPO2, ADT, NADT, etc.

Notes:

1. HR is the abbreviation for Heart Rate;
2. HRV is the abbreviation for Heart Rate Variability;
3. SPO2 stands for peripheral capillary oxygen saturation;
4. ADT is the abbreviation for Approach Detection;
5. NADT is the abbreviation for Natural Approach Detection.

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2 Hardware Introduction

2.1 Preparation

Figure 2-1 shows the components needed for evaluation. Dongle, battery and USB to Type-C cable are not provided by default.

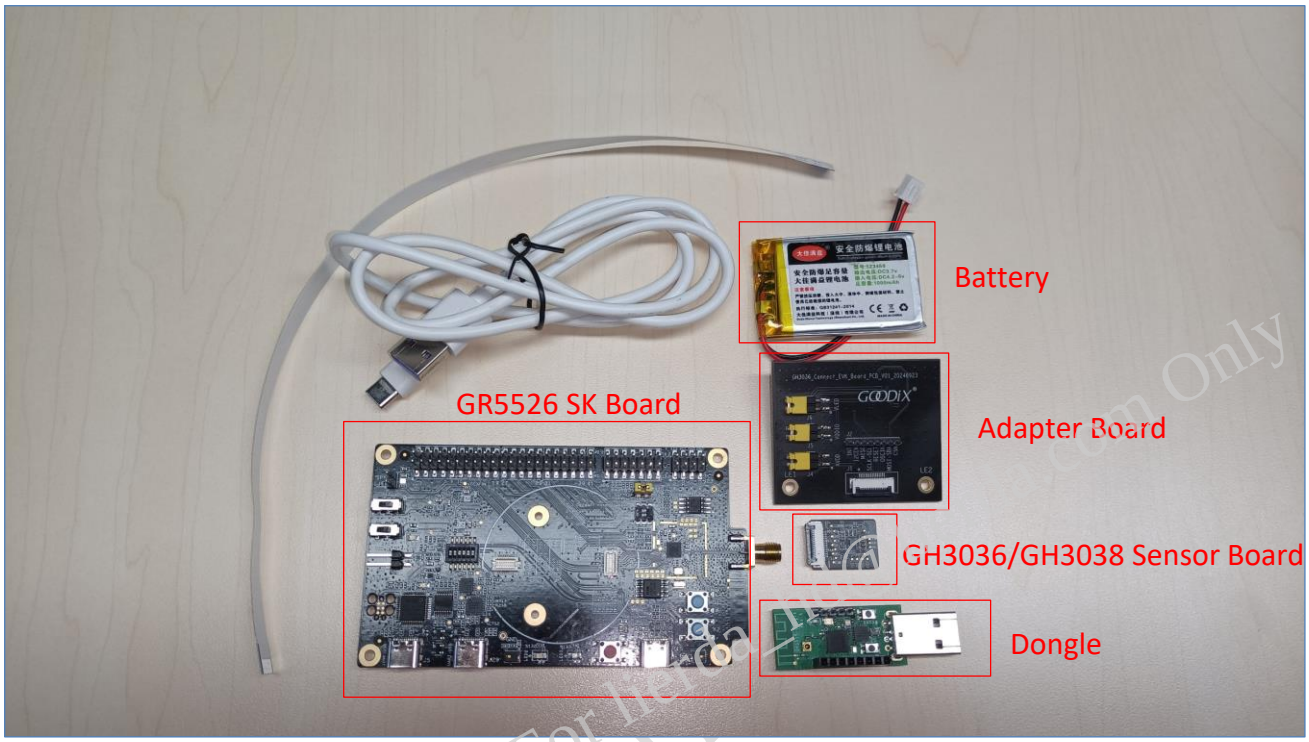

Figure 2-1 EVK components

Note:

SK Board is short for Starter Kit Board.

2.2 Hardware Connection

Take the following steps to complete the hardware connection:

1. Check the power DIP switch settings of the SK Board (see Figure 2-2);

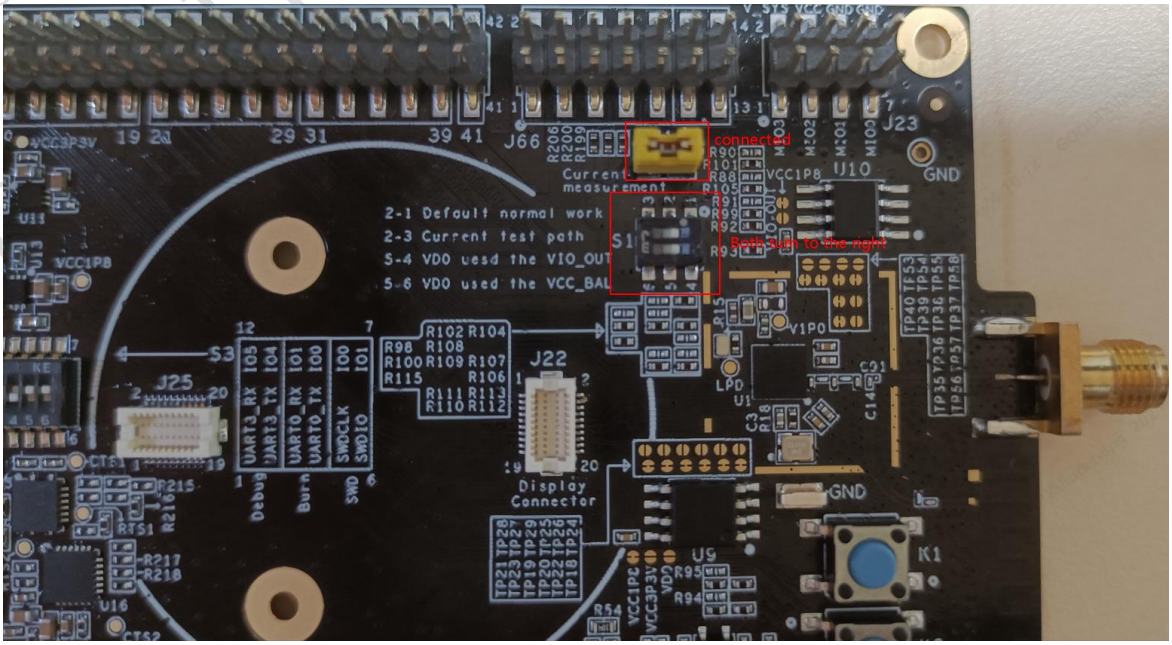

Figure 2-2 Connect SK Board to Adapter Board

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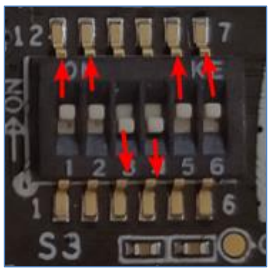
Note:

Switches on the SK Board and Adapter Board do not need to be changed, especially those in the following positions:



2. Connect the GH3036/GH3038 sensor board to the adapter board, and then connect the adapter board to the SK board (see Figure 2-2).

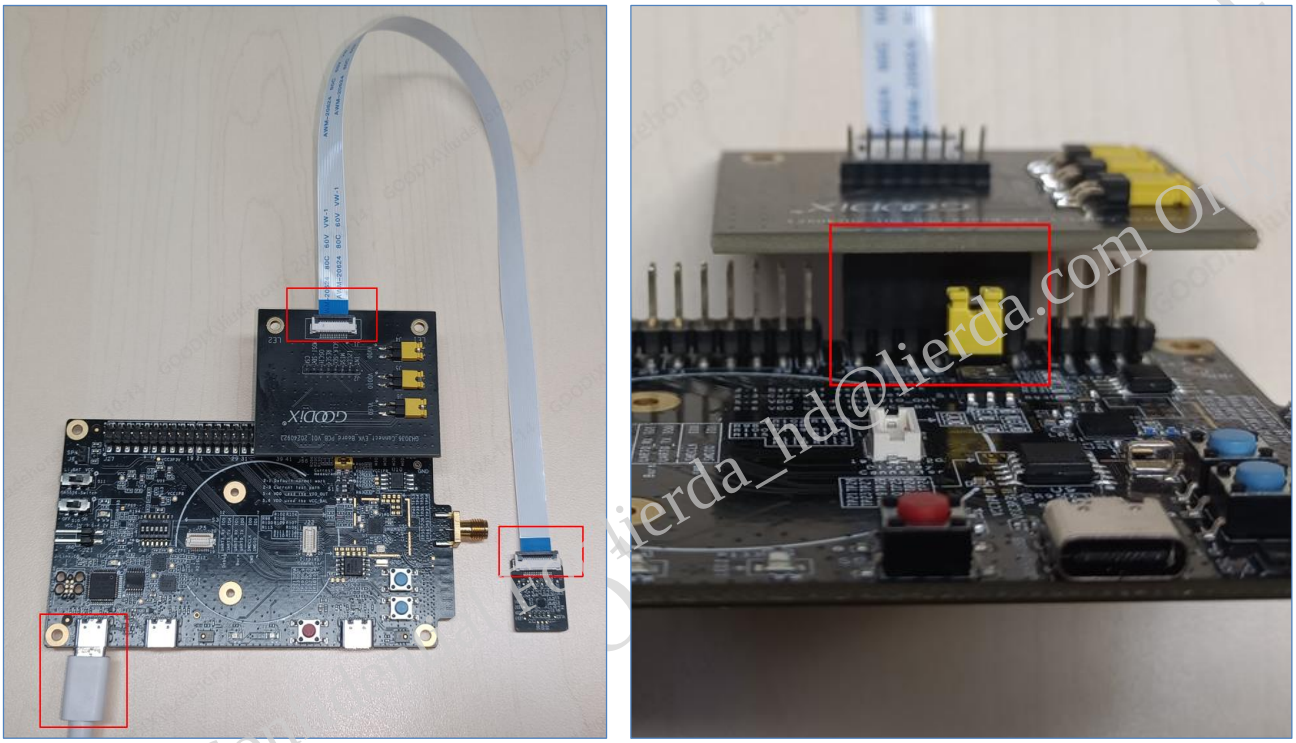


Figure 2-3 Connect SK Board to Adapter Board

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3 Software Installation

Install the GH3036&GH3038EVK_setup.exe first, and then install GH3036&GH3038_QtDLL_setup.exe.

Launch the EVK Tool installer (setup.exe), and take the following steps to install the software.

1. Click Browse and select the installation location, then click Next (see Figure 3-1);

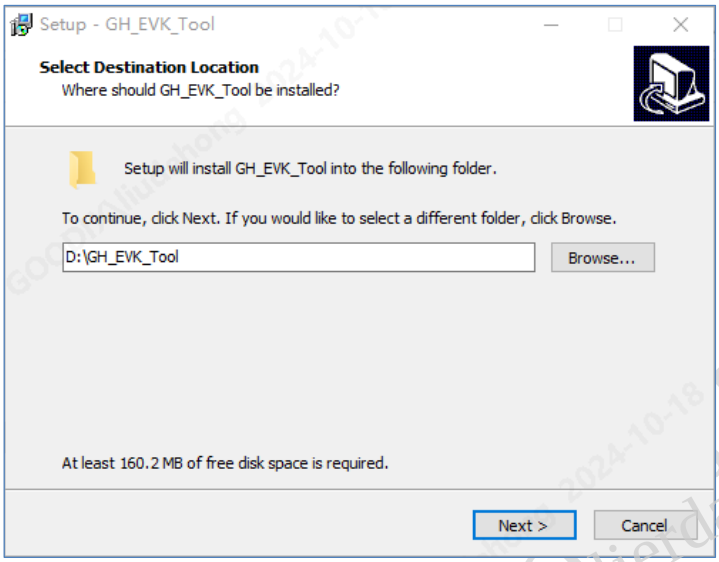

Figure 3-1 Select the installation location

2. Keep the default Start Menu Folder unchanged, then click Next (see Figure 3-2);

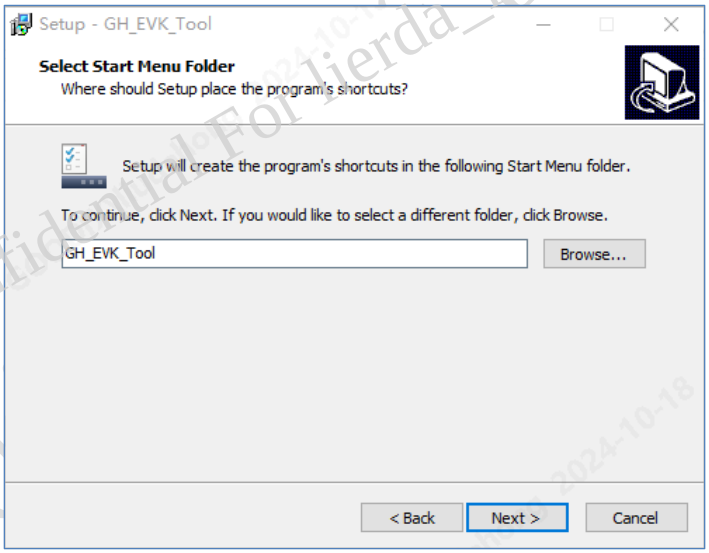

Figure 3-2 Select Start Menu Folder

3. Determine whether to create a desktop shortcut and install the drivers automatically as required and click Next (see Figure 3-3).

Note:

The first-time user is suggested to check the “Install Drivers” for automatic installation.

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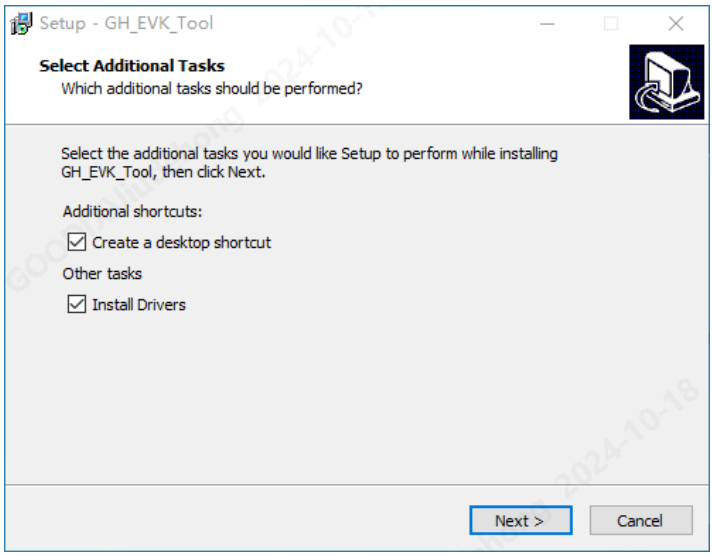

Figure 3-3 Select Additional Tasks

4. Click Install to start the installation process (see Figure 3-4).

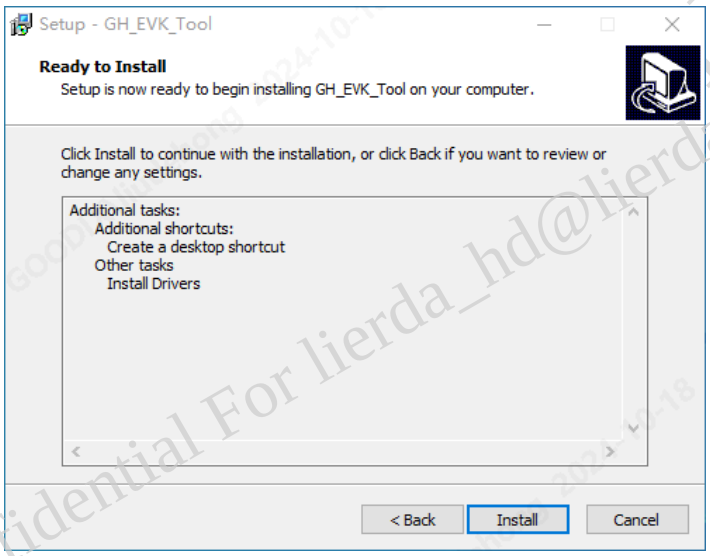

Figure 3-4 Start GH3036&GH3038EVK_setup.exe Installation

5. When the installation is complete, click Finish.

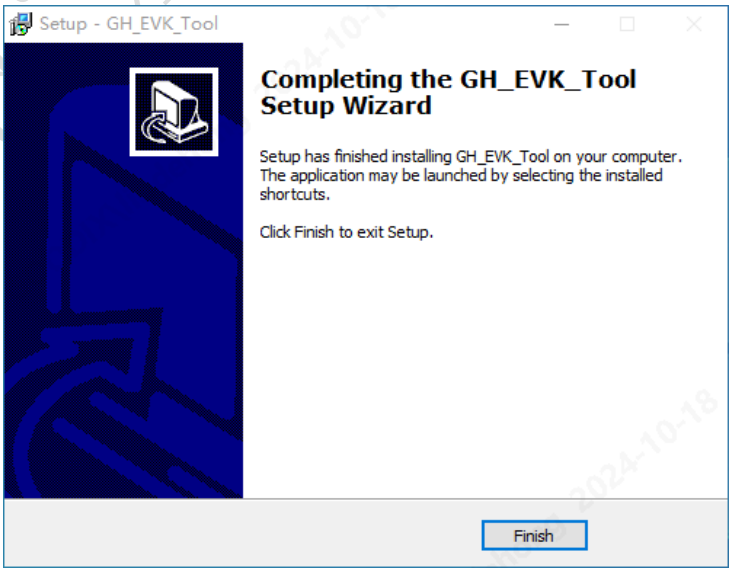

Figure 3-5 Complete GH3036&GH3038EVK_setup.exe Installation

6. Install GH3036&GH3038_QtDLL_setup.exe. Click GH3036&GH3038_QtDLL_setup.exe and install the QTDLL.

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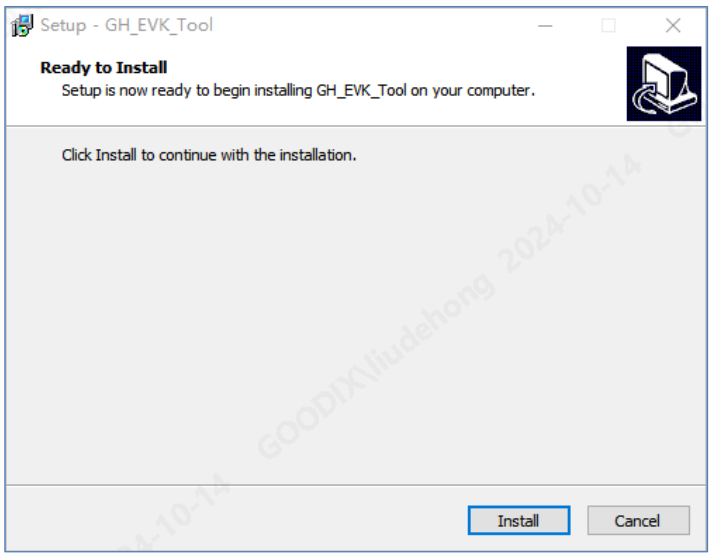

Figure 3-6 Start GH3036&GH3038_QtDLL_setup.exe. installation

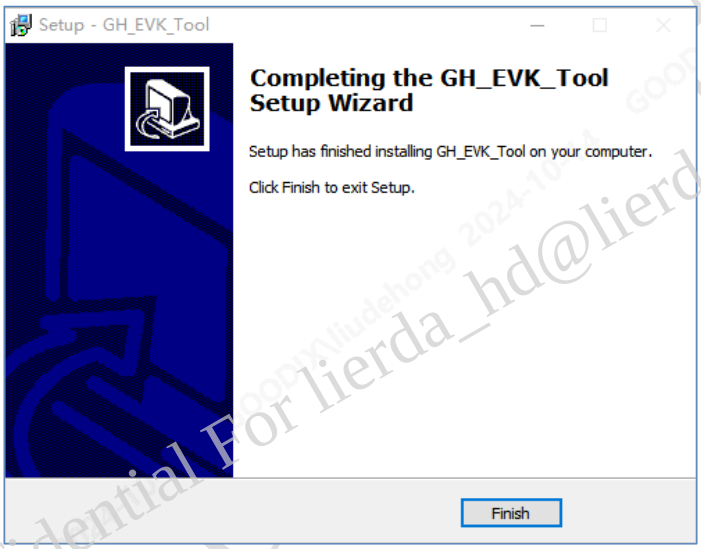

Figure 3-7 Complete GH3036&GH3038_QtDLL_setup.exe Installation

7. Open the EVK Tool (Click GHEvaluationTool or GTTestTool.exe on desktop).

Figure 3-8 EVK Tool Icon on Desktop

8. Select the chip you need, then click OK to enter the main interface of EVK Tool (see Figure 3-9).

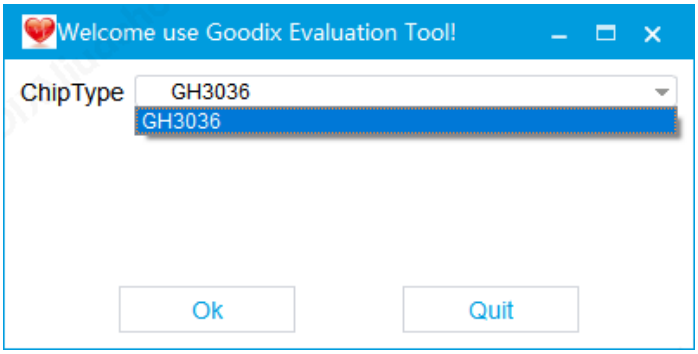

Figure 3-9 Select the chip

Note:

The EVK tool is only applicable to Windows 7 and Windows 10 operation systems.

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4 Power on the system

After preparing the hardware and installing the software, the EVK Tool is ready to be powered on.

There are two ways to connect EVK to PC. One is USB connection and the other is Bluetooth dongle connection.

4.1 Power on – Cable Connection

Connect the USB port on PC to J5 on the SK Board through Type-C cable, establishing a connection between the PC and the EVK. Switch S10 to VBAT and S11 to VCC, then the EVK Tool will be powered on. The indicator lights (LED1/LED2) on the EVK Tool will light up (see Figure 4-1).

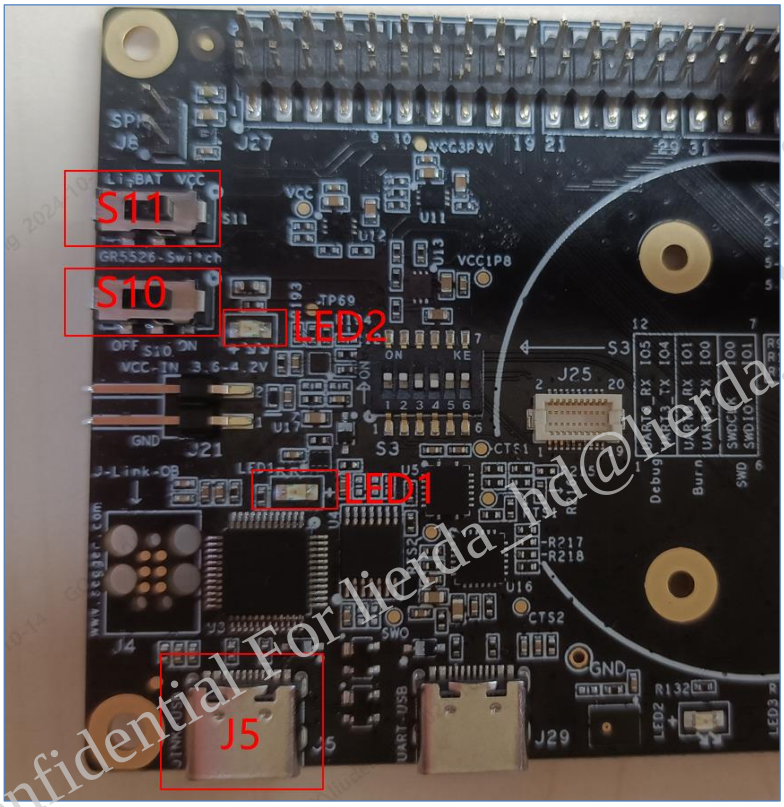

Figure 4-1 Power on the EVK after cable connection

Then You can see that the status displayed in the status bar at the lower right corner of the main interface of the EVK tool is automatically changed to "Linked to COM*xx*"(see Figure 4-2).

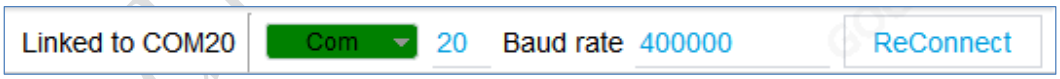

Figure 4-2 Cable connection

Note:

If the COM number cannot be directly identified, the customer can query the COM port number through the computer management, then enter the correct COM port number, and then Reconnect.

4.2 Power on – Bluetooth Connection

Plug the battery into J15 of the SK Board, and keep the Switch S10 in ON position and S11 in VCC position. After inserting the dongle to PC, the connection window of Bluetooth device will pop up (see Figure 4-3 and Figure 4-5).

If the connection window doesn't pop up, select dongle from the status bar in the lower right corner of the EVK tool, then click reconnect (see Figure 4-4).

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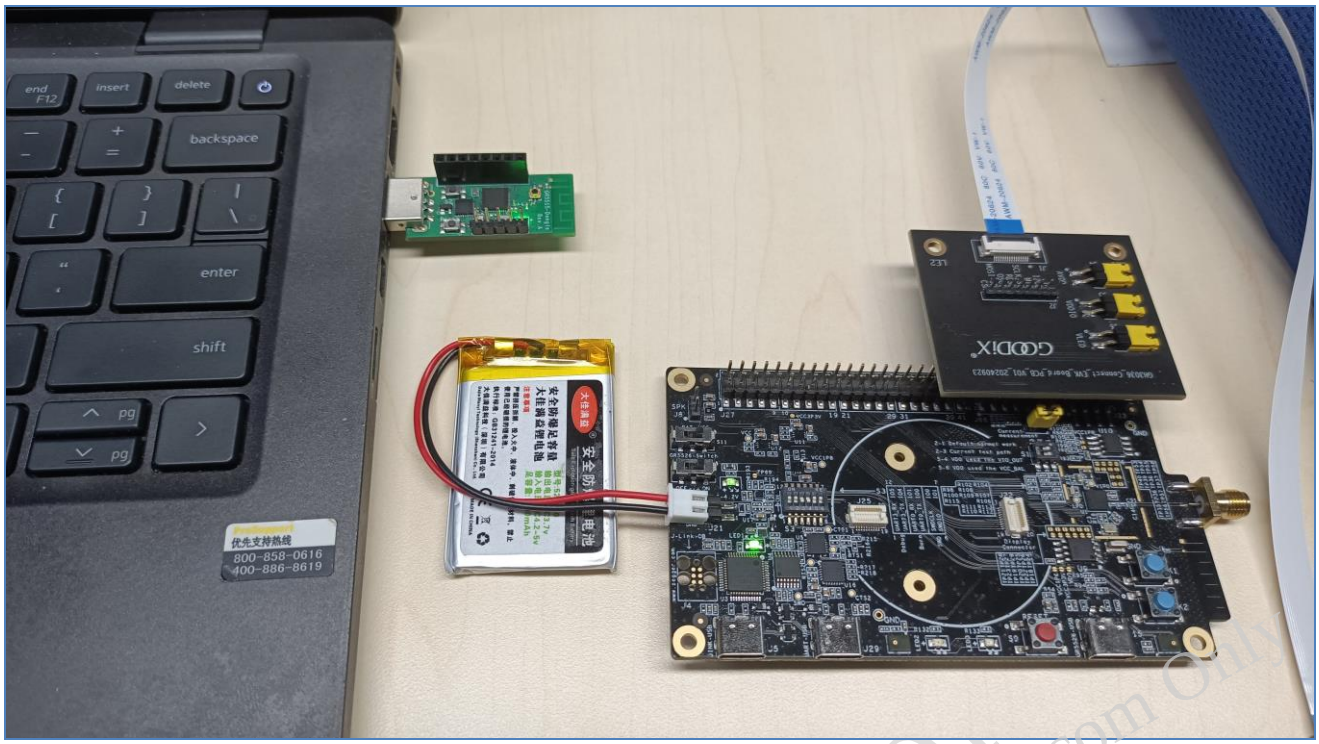

Figure 4-3 Bluetooth Connection Diagram

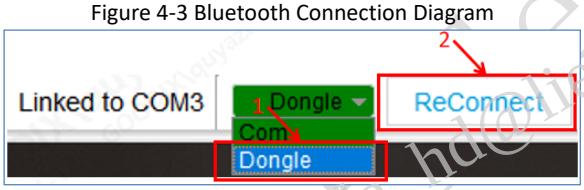

Figure 4-4 Switch the Bluetooth mode manually

Click "**Start Scan**" button to search Bluetooth devices until "GH3036_EVK" device is found; select the Bluetooth address corresponding to the device and click "Connect".

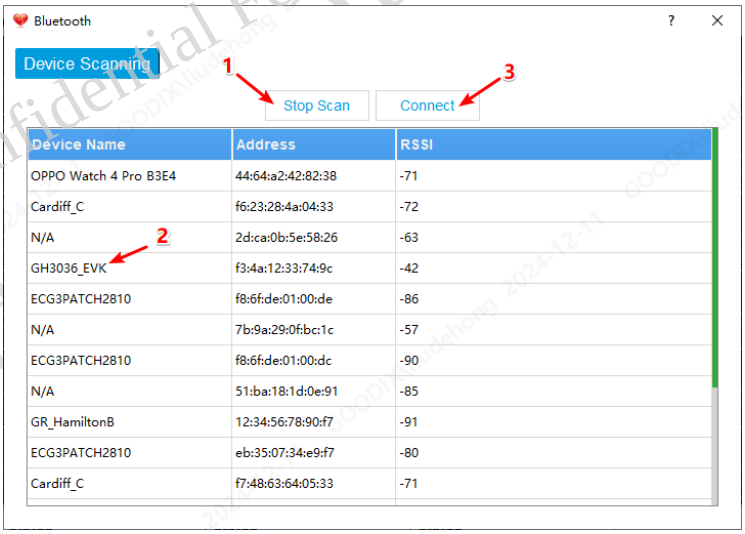

Figure 4-5 Bluetooth Connection Interface

You can see that the Bluetooth MAC address is shown at the lower right corner of the main interface of the EVK tool if connection succeeds (see Figure 4-6).

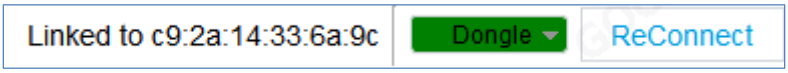

Figure 4-6 Status Displayed after Bluetooth connection succeeds

Note:

If more than one device which is named "GH3036_EVK" appears on the Bluetooth Connection Interface, select the device with the lowest RSSI value.

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5 Signal measurement

This chapter provides a brief guide on using the EVK Tool for signal measurement. The example waveforms provided below are for reference only.

5.1 PPG Measurement

Take the following steps to measure the actual PPG signal:

1. Click “Config Switch Button” to open the configuration library (see 错误!未找到引用源。);
2. Choose “PPG_3Slot_2ADCs_FS=100Hz_RF=10k_INT=316us_BGR=2.config” (see Figure 5-2), wait until the configuration is delivered, then click “Start”;
3. Data waveform will be shown in Plotter interface (see 错误!未找到引用源。).

Note:

The reference PPG configs can be found in “*installation_path\GHEvaluationTool\quick_start_config\GH3036\00-quick_started*” or “*installation_path\GHEvaluationTool\quick_start_config\GH3038\00-quick_started*”.

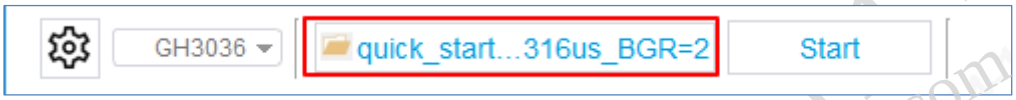

Figure 5-1 Config Switch Button

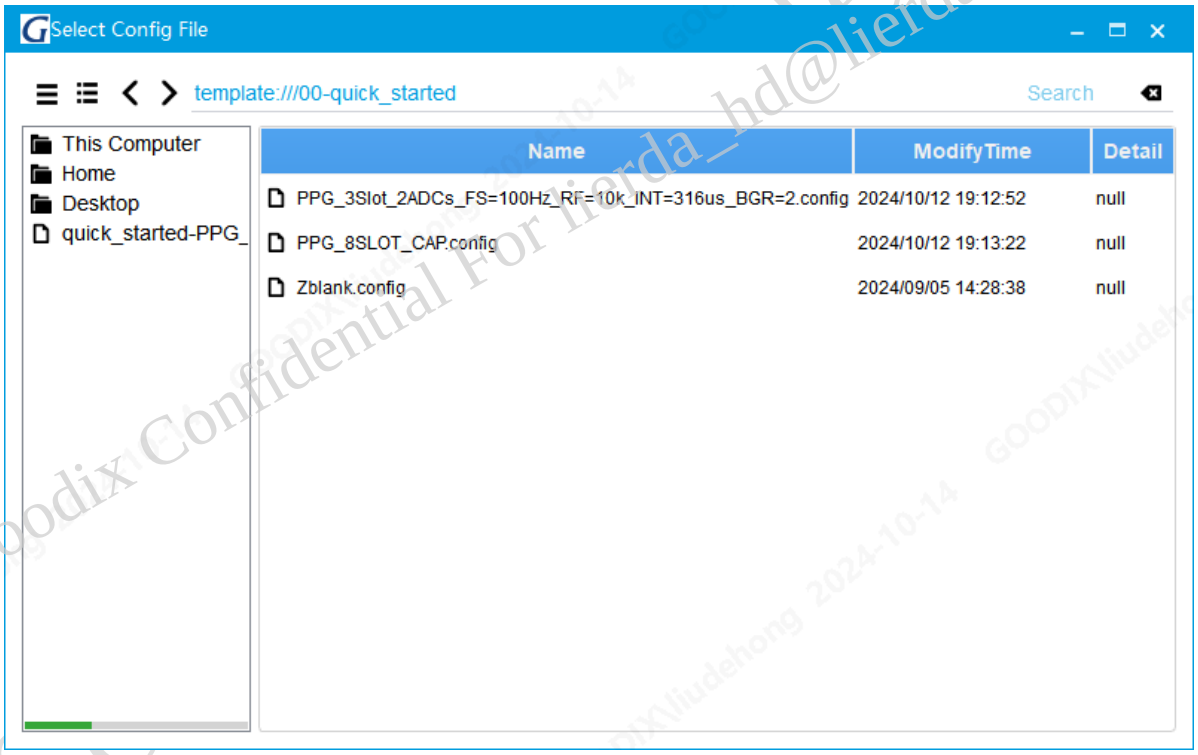

Figure 5-2 Select Config Window

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Figure 5-3 Data Waveform

5.2 PPG+CAP Measurement

Take the following steps to measure the actual PPG+CAP signal:

1. Click “Config Switch Button” to open the configuration library (see Figure 5-4);
2. Choose “PPG_8SLOT_CAP.config” (see Figure 5-5), wait until the configuration is delivered, then click “Start”;
3. Data waveform will be shown in Plotter interface (see Figure 5-6).

Note:

The reference configuration can be found in “*installation_path\GHEvaluationTool\quick_start_config\GH3036\02-application_ref*” or “*installation_path\GHEvaluationTool\quick_start_config\GH3038\02-application_ref*”.

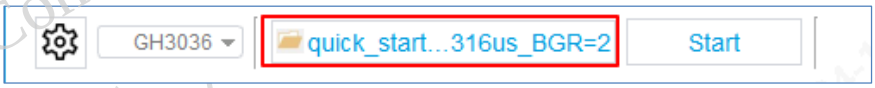

Figure 5-4 Config Switch Button

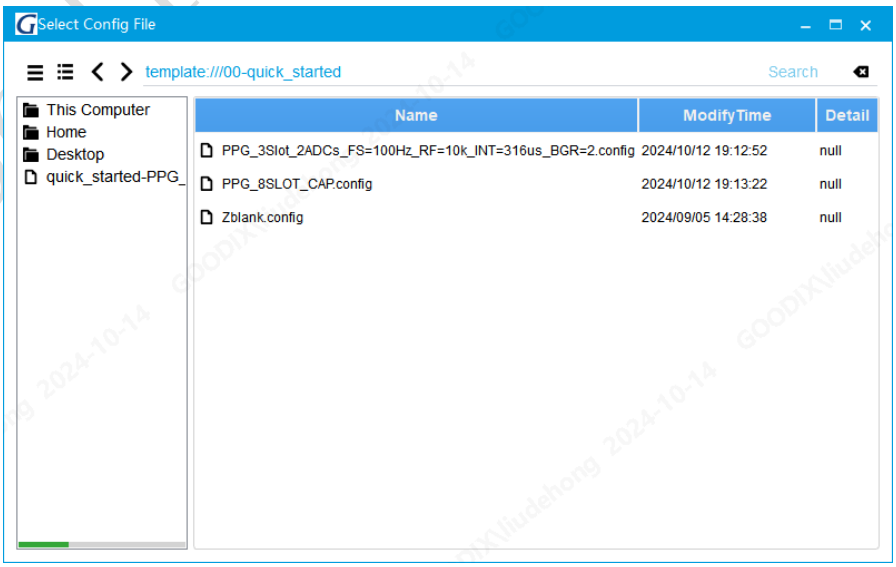

Figure 5-5 Select Config Window

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Figure 5-6 PPG+CAP Data&Result Waveform

5.3 HR+ADT Measurement

Take the following steps to enable HR+ADT functions:

1. Click “Config Switch Button” to open the configuration library (see Figure 5-7);
2. Choose “HR_ADT.config” (see Figure 5-8), wait until the configuration is delivered, then click “Start”;
3. Data waveform will be shown in Plotter interface (see Figure 5-9).

Note:

The reference configuration can be found in *“installation_path\GHEvaluationTool\quick_start_config\GH3036\02-application_ref”* or *“installation_path\GHEvaluationTool\quick_start_config\GH3038\02-application_ref”* .

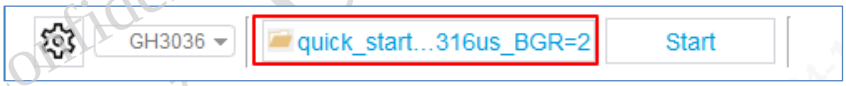

Figure 5-7 Config Switch Button

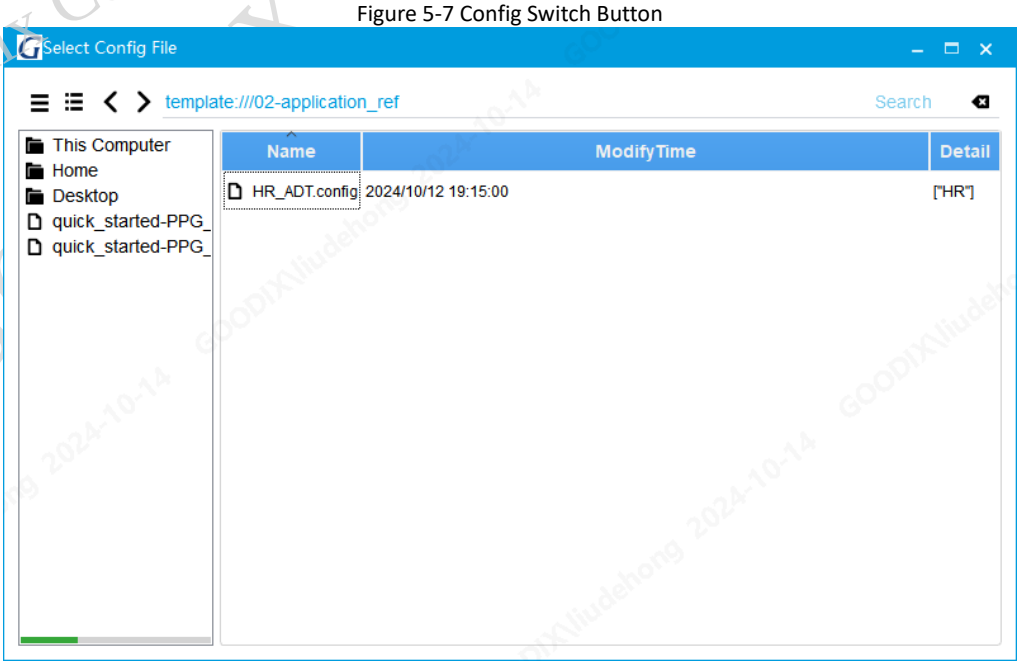

Figure 5-8 Select Config Window

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Figure 5-9 ADT+HR Data&Result Waveform

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6 Check the raw data

The collected raw data is automatically saved in "*installation path*\ GHEvaluationTool \ADC_DATA" (see 错误!未找到引用源。).

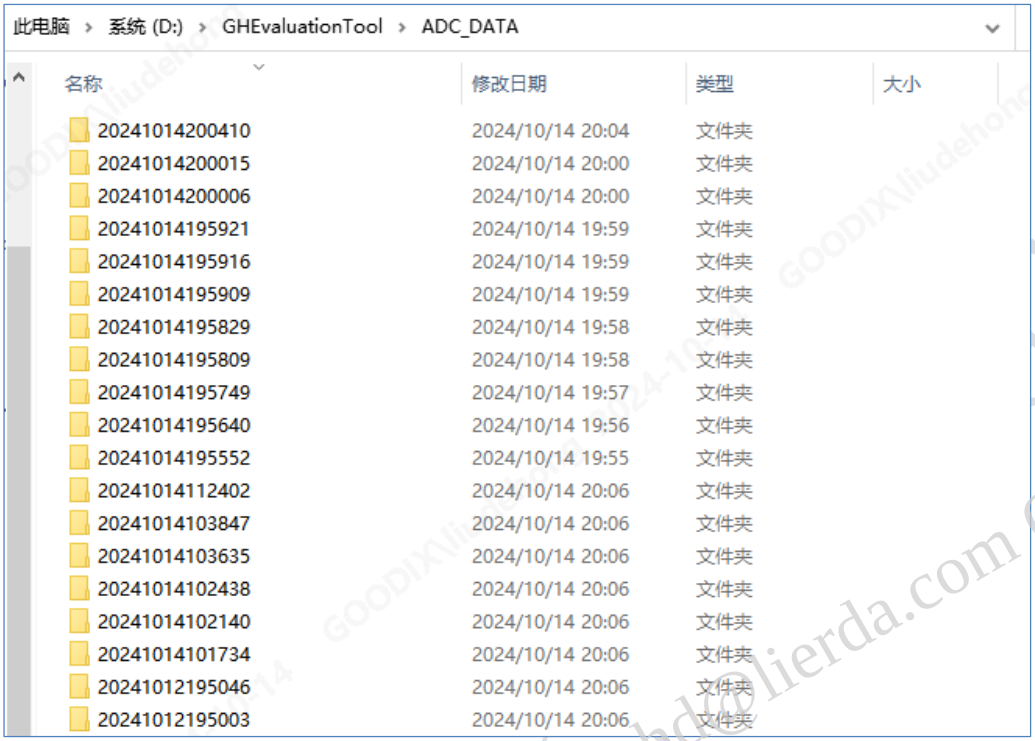

Figure 6-1 Raw data files path

The raw data is saved in a CSV file, and the main information is as below:

- Time stamp: Unix time stamp of sample
- FRAME_ID: Increased from 0 to 255; the frame ID is useful to check the packet loss
- Data:
  - ACCX\ACCY\ACCZ: G-sensor data
  - Ipd: PD current, unit: pA
  - Flag: Refer to app-format
  - Ref_result: Reference algorithm result(optional)
  - Algo_result: Algorithm result(optional)
  - Rawdata: PPG rawdata
  - AGC_INFO: Auto gain control information
  - LED_INFO: Refer to app-format
  - GYRO_X\GYRO_Y\GYRO_Z: 9. Gyroscope data

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